UNIVERSITY OF OXFORD

DEPARTMENT OF PAEDIATRICS

METHODOLOGY, EVIDENCE SYNTHESIS, AND IMPLEMENTATION | MESISIG
IASP Special Interest Group

# Navigating AI in Evidence Synthesis

## AI tools across the review workflow

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Map and presentation

lukebax.github.io/ai-review-tools/

# Stages of a systematic review

1 Protocol

2 Search

3 Screening for relevance

4 Quality assessment

5 Data extraction

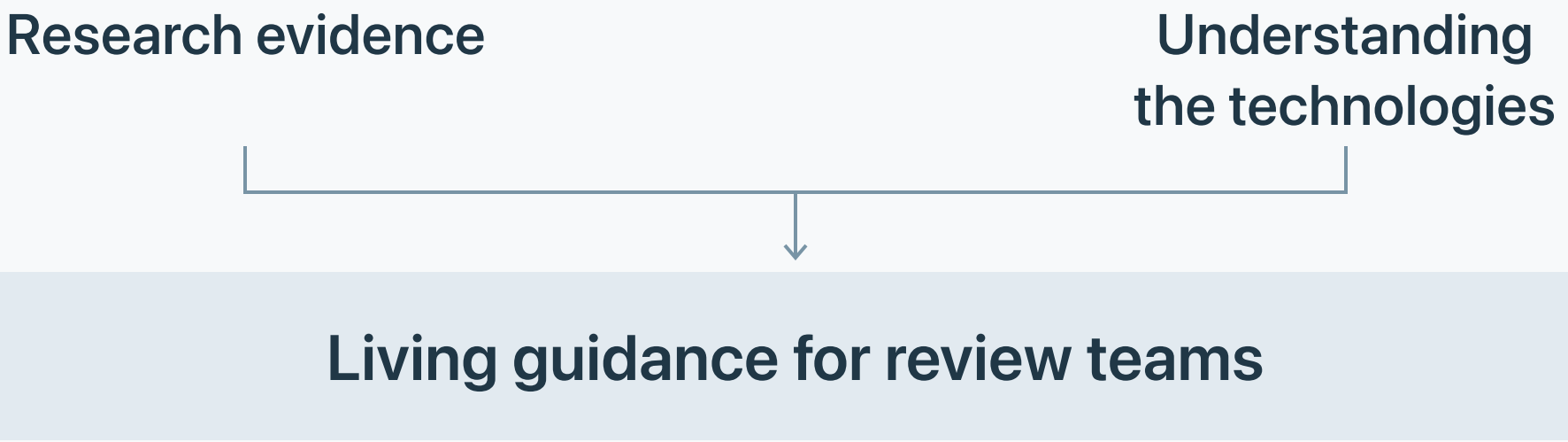
6 Data synthesis

7 Certainty of evidence

8 Manuscript

# RAISE provides guidance for using AI in reviews

Responsible use of AI in evidence synthesis

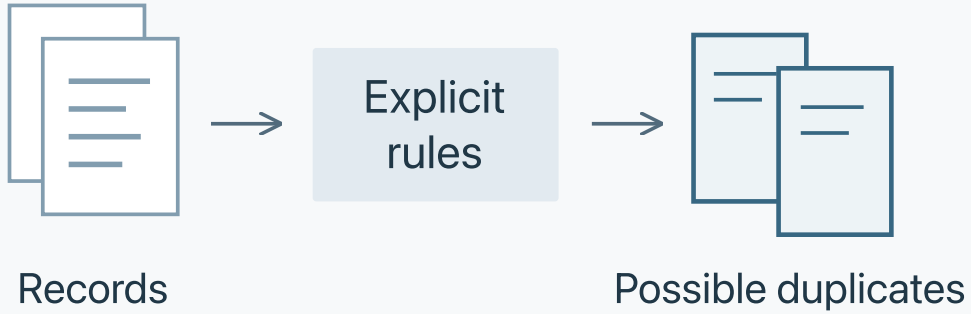



Developed by researchers working in evidence synthesis

# Explicit rules and patterns in unlabelled data

## Rule-based algorithms

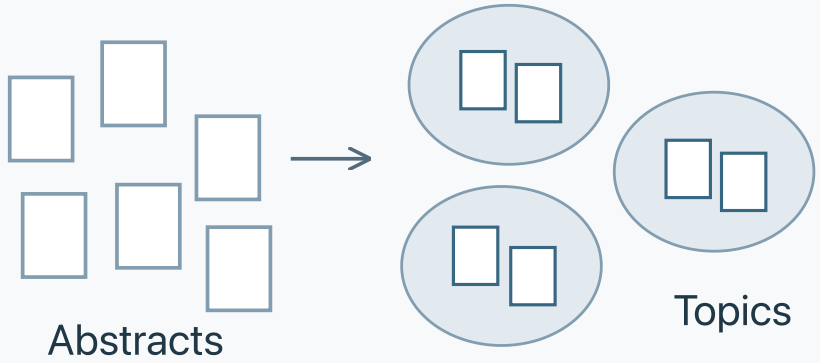
Deduplication




Predefined rules compare the records.

## Unsupervised machine learning

Topic modelling



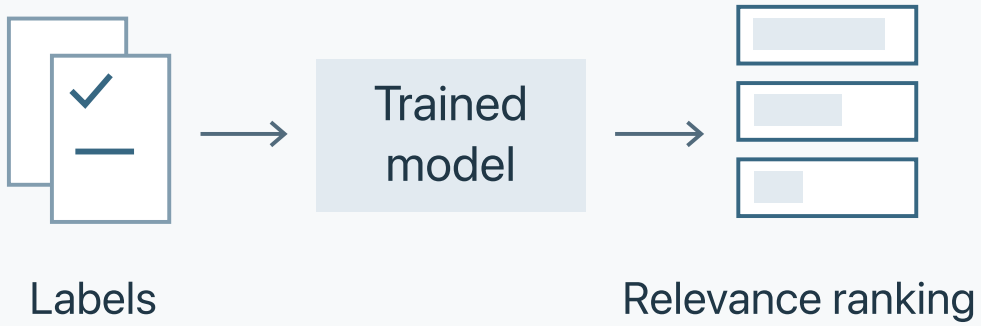

Patterns emerge without supplied labels.

These are broad, overlapping categories; individual tools can combine approaches.

# Learning from labelled examples and generating language

## Supervised machine learning

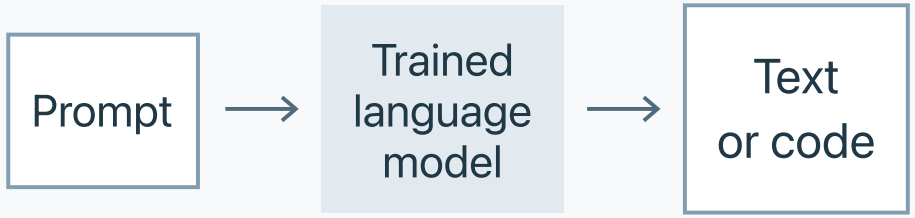
Prioritised screening




Already trained or trained during the review

## Generative LLMs

Writing analysis code

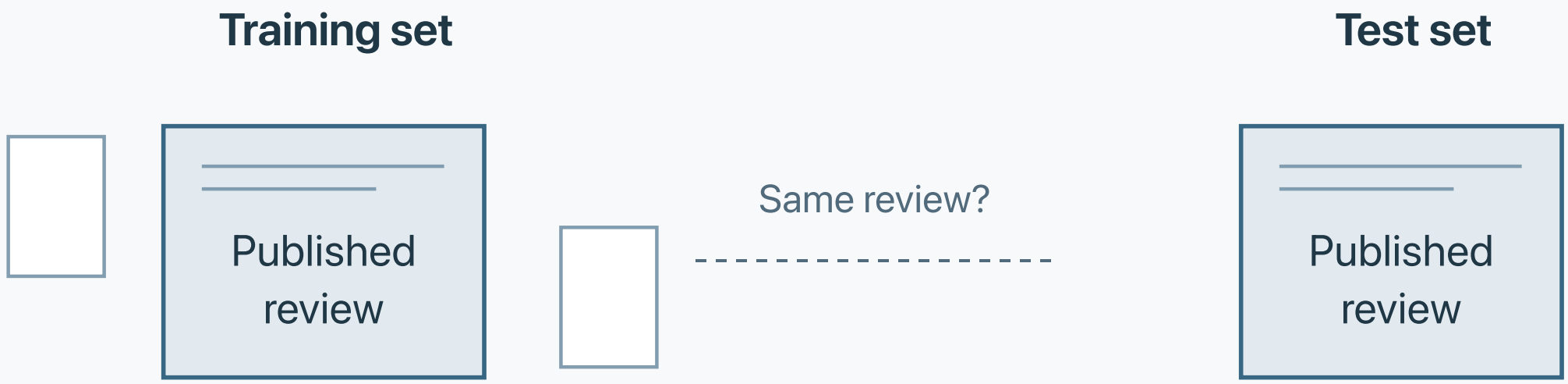



Outputs can vary.
Models can hallucinate.

These are broad, overlapping categories; individual tools can combine approaches.

# Evaluation needs independence from development

Possible training–test overlap: a hypothetical example




Apparent performance may overstate performance on new material.

Incomplete disclosure makes overlap harder to establish.

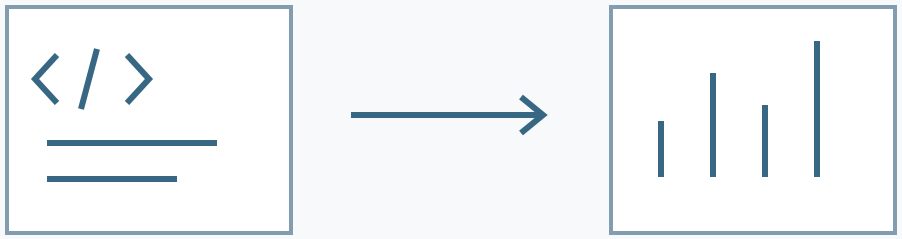
# Recommendations apply to a particular use

| | |
|---|---|
| **Acceptable for use** | Use outputs, accounting for limitations and bias |
| **Human verification required** | Check outputs and correct errors |
| **Requires validation within the review** | Test performance against a prespecified threshold and suitable benchmark |
| **Exploratory and supplementary use** | Explore or supplement established methods |
| **Not acceptable for use** | Do not rely on outputs for this task |

# Verification and validation ask different questions

## Verification

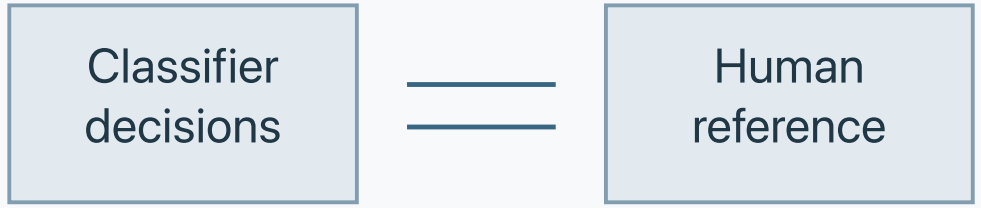
Check the output




Does it carry out the intended analysis?

## Validation

Formally evaluate performance



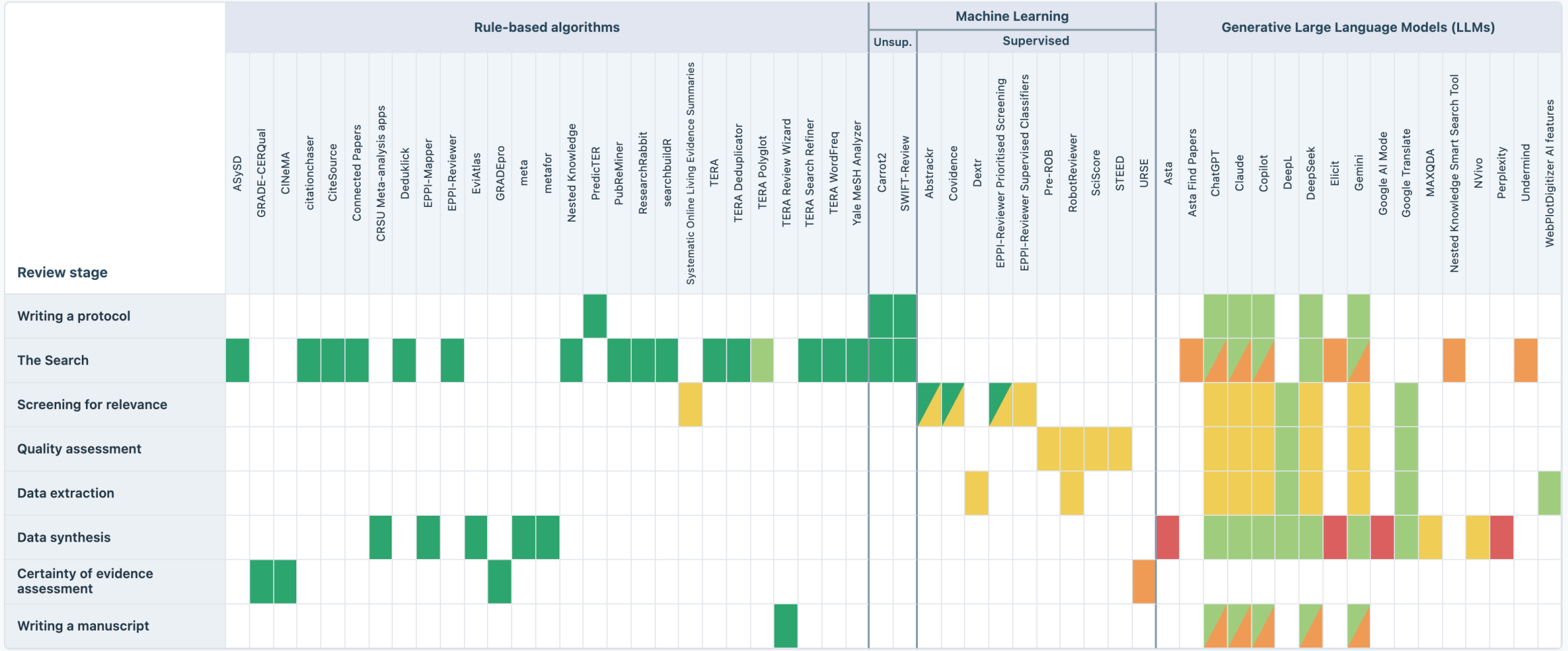

Do relevance judgements meet the prespecified standard?

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Use robust, relevant evidence. Follow any requirement for validation within the review.

A workflow may need both.

# AI tools across the review stages






Find interactive map here: lukebax.github.io/ai-review-tools/

# Reading the patterns in the map

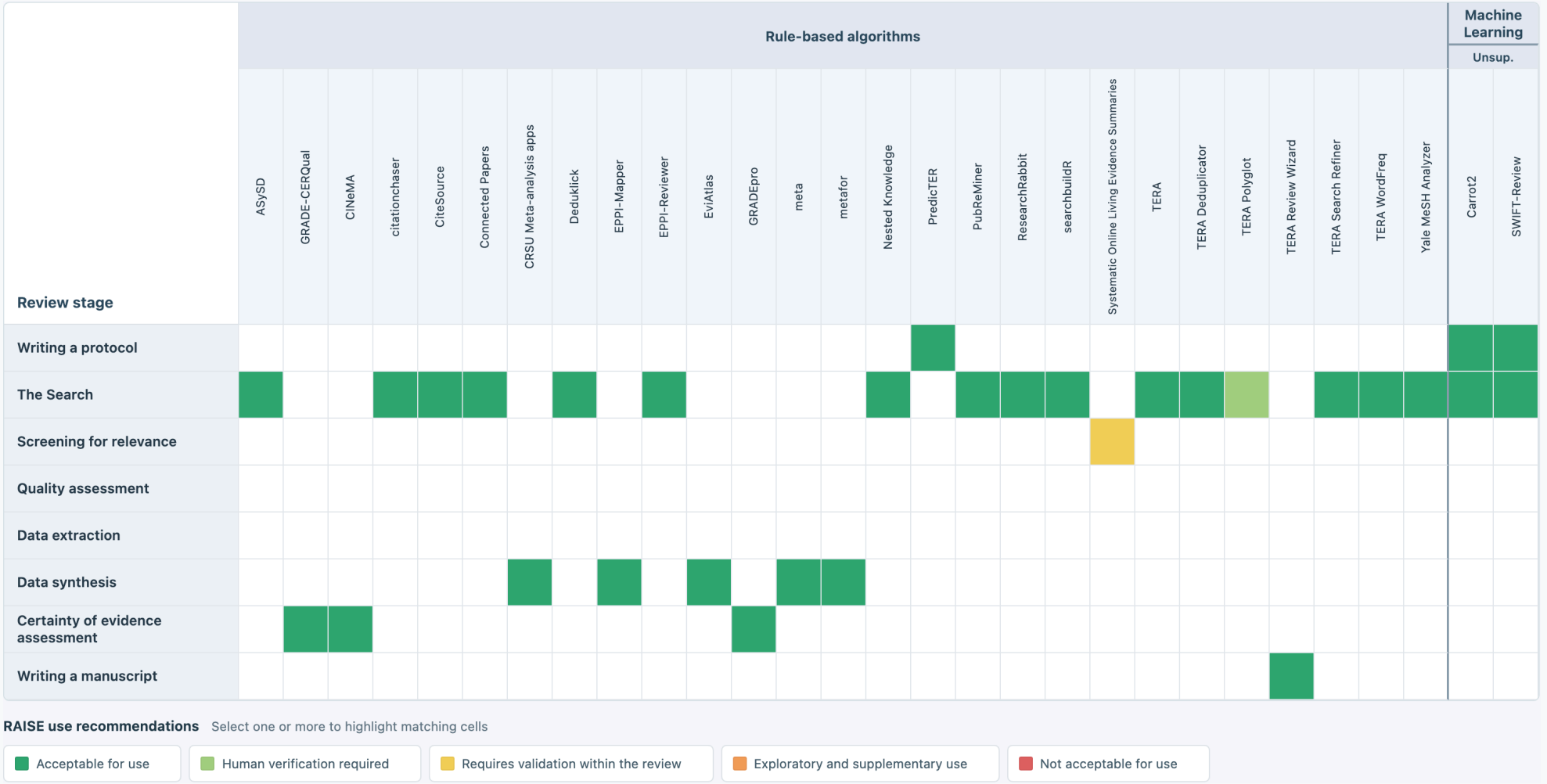
Selected tools and tasks from RAISE's guidance

Tasks, methods, maturity and supporting evidence vary together.

The patterns do not independently establish that one AI family is safer.

# Rule-based algorithms and unsupervised machine learning

## Most of these mapped uses are acceptable



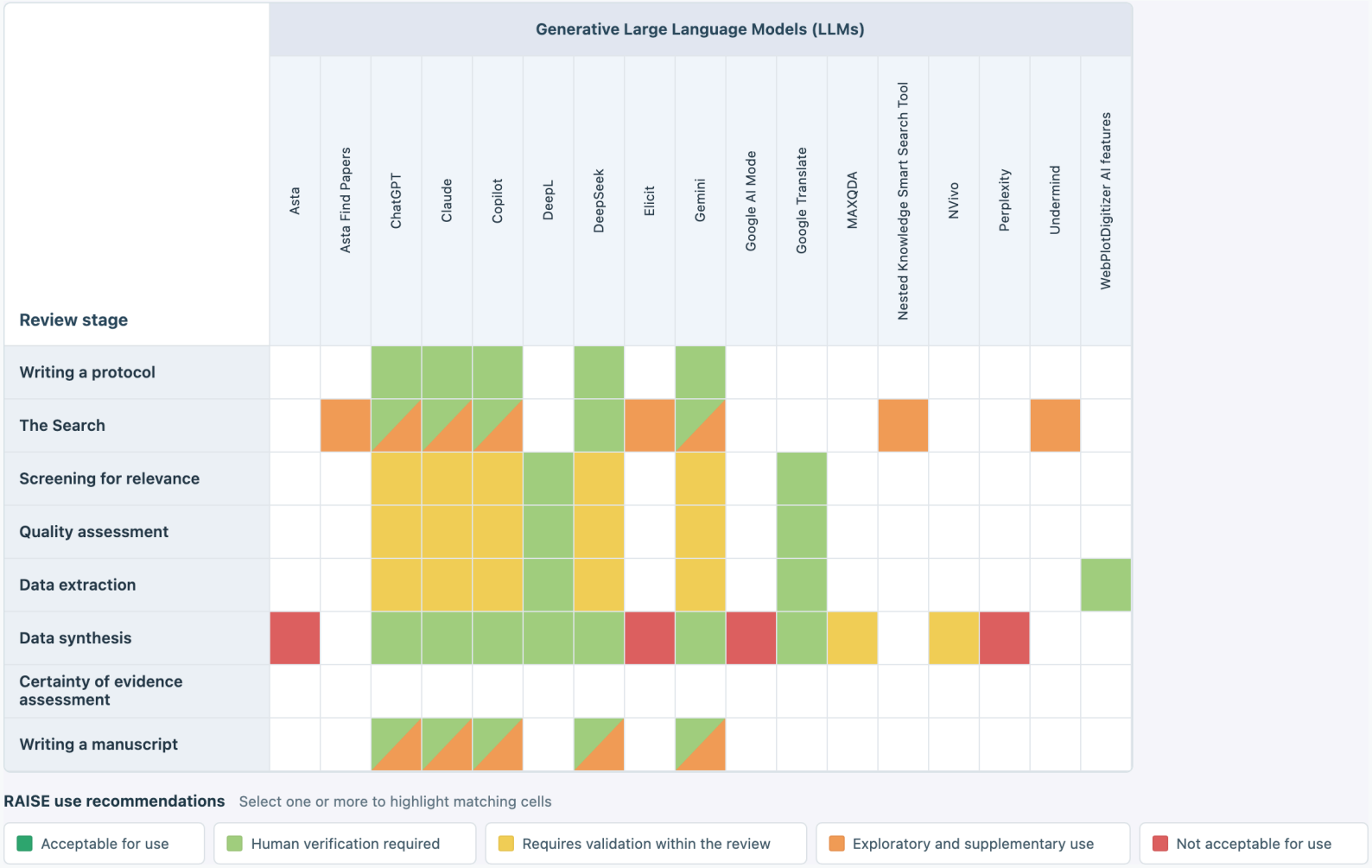
# Supervised machine learning

## Most mapped uses need validation: performance depends on task and material



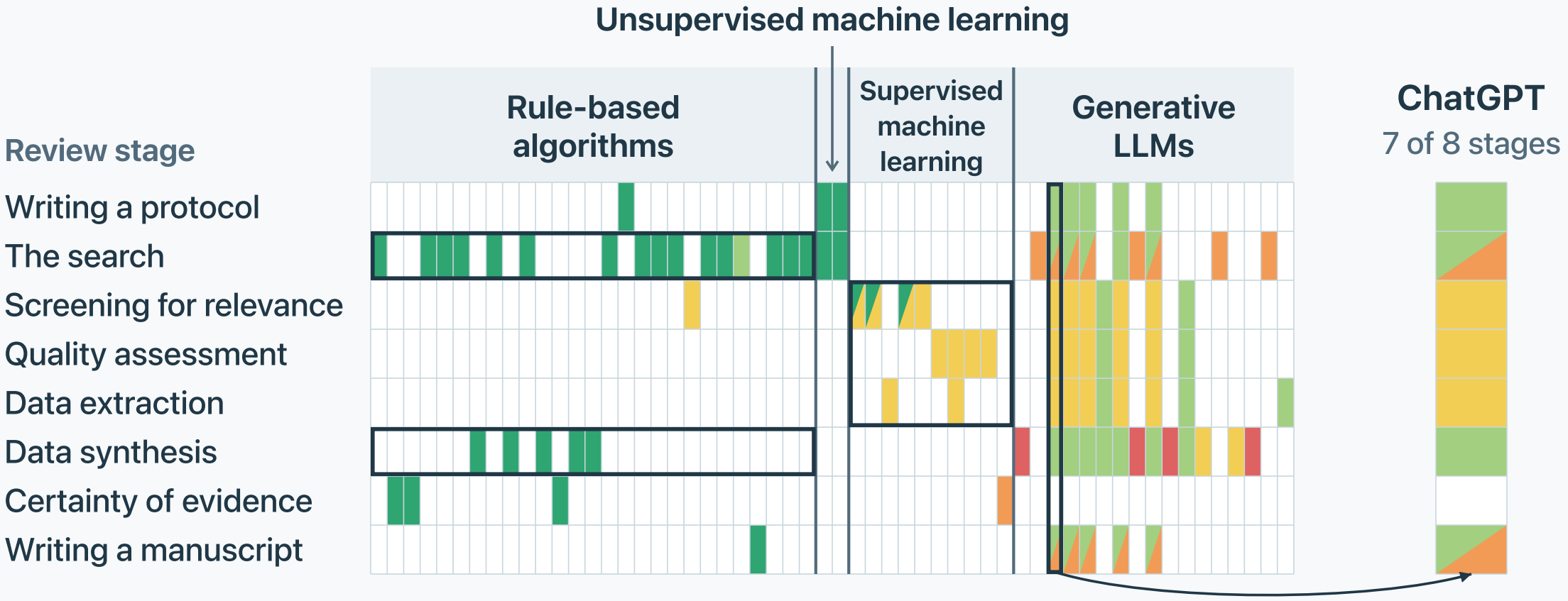
# Generative large language models (LLMs)

Recommendations depend on the task, evidence and safeguards.



# Tools differ in where and how broadly they are used

Stage clusters in the overview, with ChatGPT enlarged alongside




Rule-based
Record matching and calculations

Supervised
Relevance, methods and data

ChatGPT
Language and code across stages

# Prioritised screening with and without early stopping

Stopping early changes reliance on the ranking

**Abstrackr**



**Covidence**



**EPPI-Reviewer Prioritised Screening**



## Screen every record

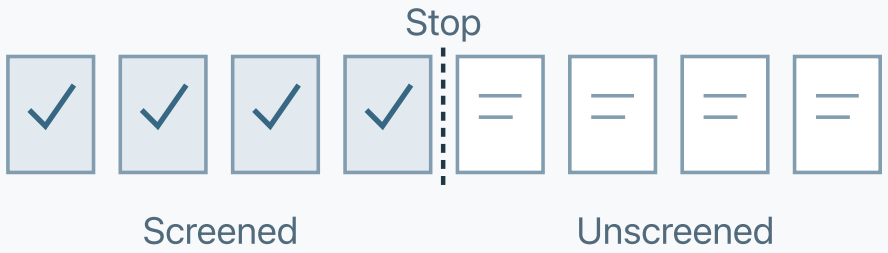


Every record is screened

**Acceptable for use**

A poor ranking does not leave records unseen.

## Stop early




**Requires validation within the review**

Assess the screening and stopping procedure.

The ranking can be identical. What changes is how much we rely on it.

# Two different uses during screening

Screening for relevance

## Google Translate

Translation support

**Human verification required**

Check the translation against the original.

## Google’s Gemini

Relevance judgements

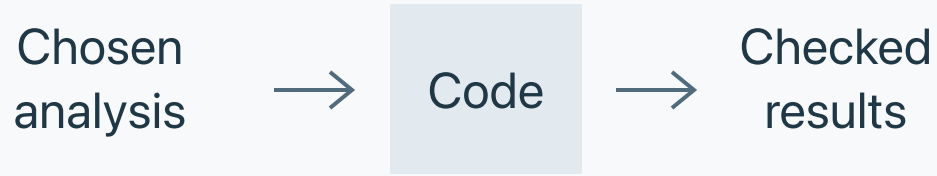
**Requires validation within the review**

Evaluate the screening procedure’s performance.

Selected screening cells from the map.

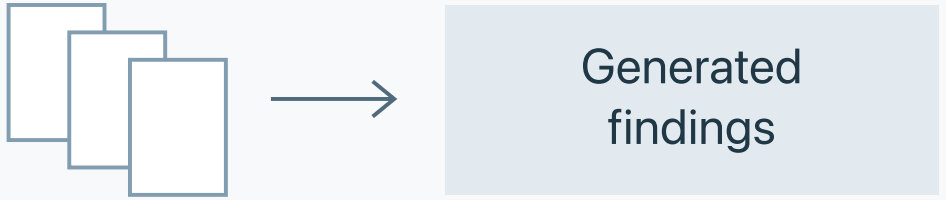
# Synthesising findings differs from drafting analysis code

## LLM-drafted code for a chosen analysis




**Human verification required**

## LLM-derived synthesis across studies




**Not acceptable for use in a formal review**

Qualitative coding or themes: validation and verification.
The boundary with cross-study synthesis is not fully defined.

# RAISE also restricts complete agentic reviews

**Complete agentic review: not acceptable for use**




Using an LLM agent for one review task requires its own assessment.

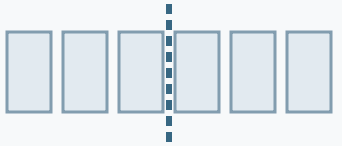
# Tool suitability

Prioritised screening with early stopping

1

## What is the purpose of the tool?

Assess ranking plus the stopping procedure.



2

## Where have the training, testing and validation data come from?

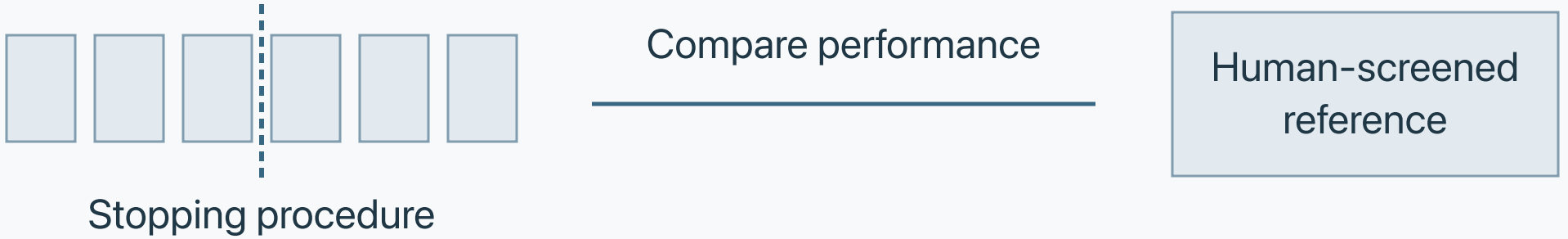
Training labels: quality and selection.
Testing and validation data: independence and relevance.



# Validation and performance

Prioritised screening with early stopping

## 3 Is the tool validated and performing sufficiently well for use?




Performance of the stopping procedure

Set the performance standard in advance.
Establish whether it is met within this review.

An appropriate human-screened reference

# Team, resources and documentation

Prioritised screening with early stopping

## 4 Does the team have the expertise and resource to use the AI tool effectively, and is the usability acceptable?

Skills, assessment time and a workable process

## 5 Is there sufficient transparency around licenses and other available documentation?

Instructions, access, data handling, version and settings

Missing stopping evidence must be resolved before reliance.

# Choosing a suitable use of AI

Consider how the tool works, the task and the evidence needed before relying on its outputs.



RAISE provides the guidance.

The companion map helps you explore RAISE's recommendations and the tools.

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Map and presentation

**lukebax.github.io/ai-review-tools/**